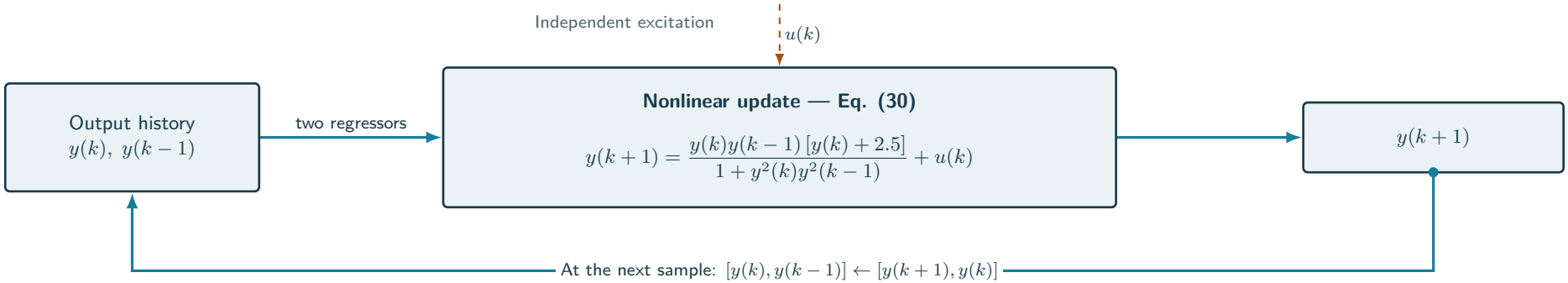
# Artificial dynamic systems: two exact recurrences

Paper section 3.1: *Artificial system identification problems*

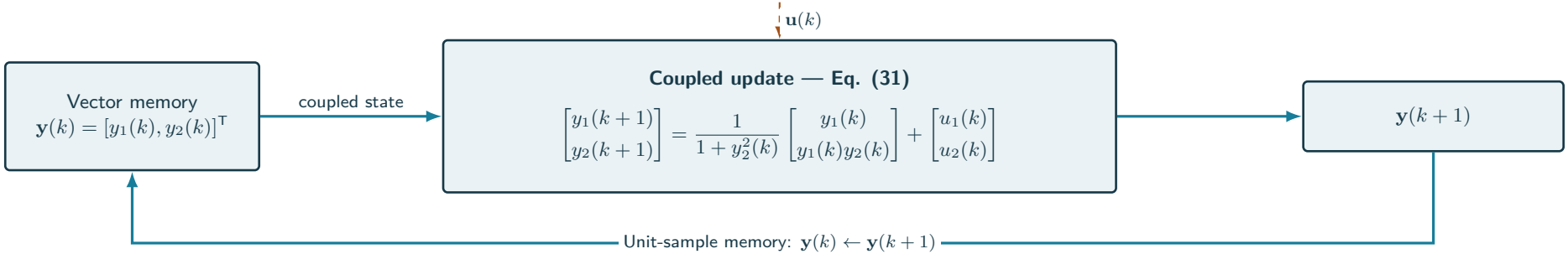
## A. SISO: Narendra–Parthasarathy example 2

One input, one output; second-order recurrence




## B. MIMO: Narendra–Parthasarathy example 4

Two inputs, two coupled outputs




**Identification excitation**

Article: $10^5$ i.i.d. samples.
SISO: $u \sim \mathcal{U}[-2, 2]$.
MIMO: each $u_i \sim \mathcal{U}[-1, 1]$.

**Validation excitation**

SISO: $u(k) = \sin(2\pi k/25)$.
MIMO: $u_1(k) = \sin(2\pi k/25)$, $u_2(k) = \cos(2\pi k/25)$.

**Evaluation contract**

Distinguish measured-history one-step prediction from recursive free-run prediction. Declare initialization, sample count and seed.

Blue arrows: numerical state/history propagation. Dashed orange arrows: independent excitation. These artificial identification problems have no implied physical plant. Repository zero-history initialization and random seeds are declared experimental choices.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

# Data resources: from records to a reproducible benchmark

Paper section 3.2: *Data repositories for system identification and regression*

## Seven resource families named in the article

**DaISy**
Dynamic identification records

**IFAC**
Identification benchmark records

**NIST**
Statistical reference data

**MATLAB**
Toolbox and demonstration data

**UCI**
Machine-learning data repository

**DELVE**
Learning-method evaluation data

**WEKA**
Benchmark data collections

These are sources of experiments and regression tasks, not seven physical process models. Availability, licensing and original links must be verified for each selected record.

One selected source record

**1. Acquire + identify**
Source URL, license, version, checksum and apparatus description

→

**2. Import + inspect**
Time base, input/output order, units, missing values and operating range

→

**3. Freeze the split**
Separate fitting, selection and final evaluation records before estimation

→

**4. Compare models**
Same samples, excitation, horizon, metrics and uncertainty conventions

**Dynamic identification**

Keep temporal order and independent runs. Prevent overlap of lagged regressors across split boundaries; state the sample period and prediction horizon.

**Static regression**

State the target, feature units and sampling unit. Preserve grouped dependence. Use a declared validation split and a final held-out test.

**Shared safeguards**

Fit normalization only on training data. Record exclusions, provenance and random seeds. Report baseline errors with the same evaluation contract.

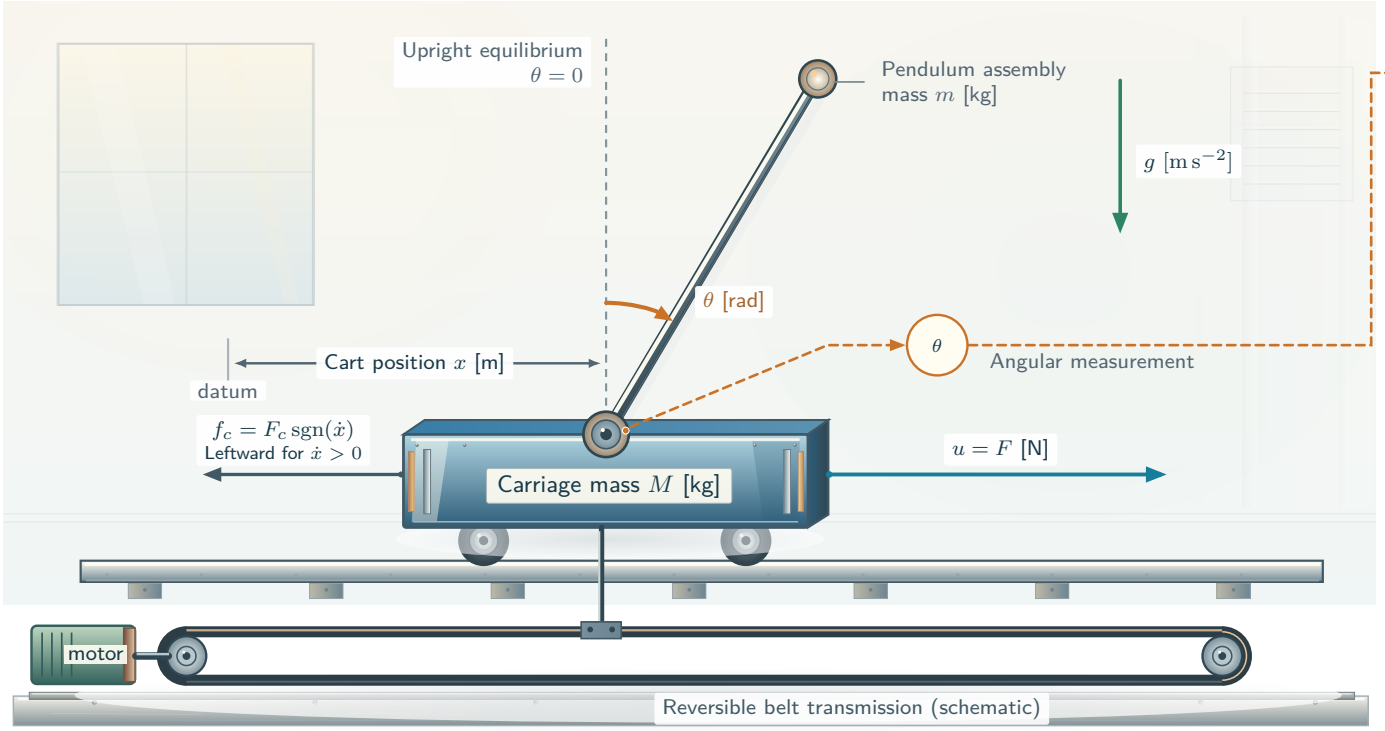
This resource map explains Section 3.2 and the repository import workflow. It does not claim that every historical third-party data set is bundled or still downloadable. Dataset-specific provenance remains necessary before numerical comparison.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

# Cart and inverted pendulum

Paper section 3.3: *Cart with inverted pendulum*

## Belt-driven carriage and freely rotating pendulum




**Control problem**

Swing up from the hanging position, balance near the unstable upright state, then position the cart.

**Published measurement**

Angular position $y = \theta$.

**Simulation state order**

$[\theta,\ \dot{\theta},\ x,\ \dot{x}]^{\mathsf{T}}$

Units: rad, rad/s, m, m/s.

**Nonlinear couplings**

Gravity $\sin\theta$, inertial terms $\dot{\theta}^2$, trigonometric coupling and Coulomb friction.

Angular joint loss: $d_{Mf}\dot{\theta}$ [N m].

**Input and actuator limit**

$u = F$ [N], $|F| \le F_{\max}$.

The source uses a motor and belt; Eq. (32) takes the resulting force as its input.

**Length convention**

$l$ [m] is the characteristic length used in Eq. (32). The drawing does not infer a separate rod length, inertia or centre of mass from $l$.

**Reading the drawing**

Solid arrows: forces / motion.

Dashed arrows: measurement signals.

Geometry is not to scale.

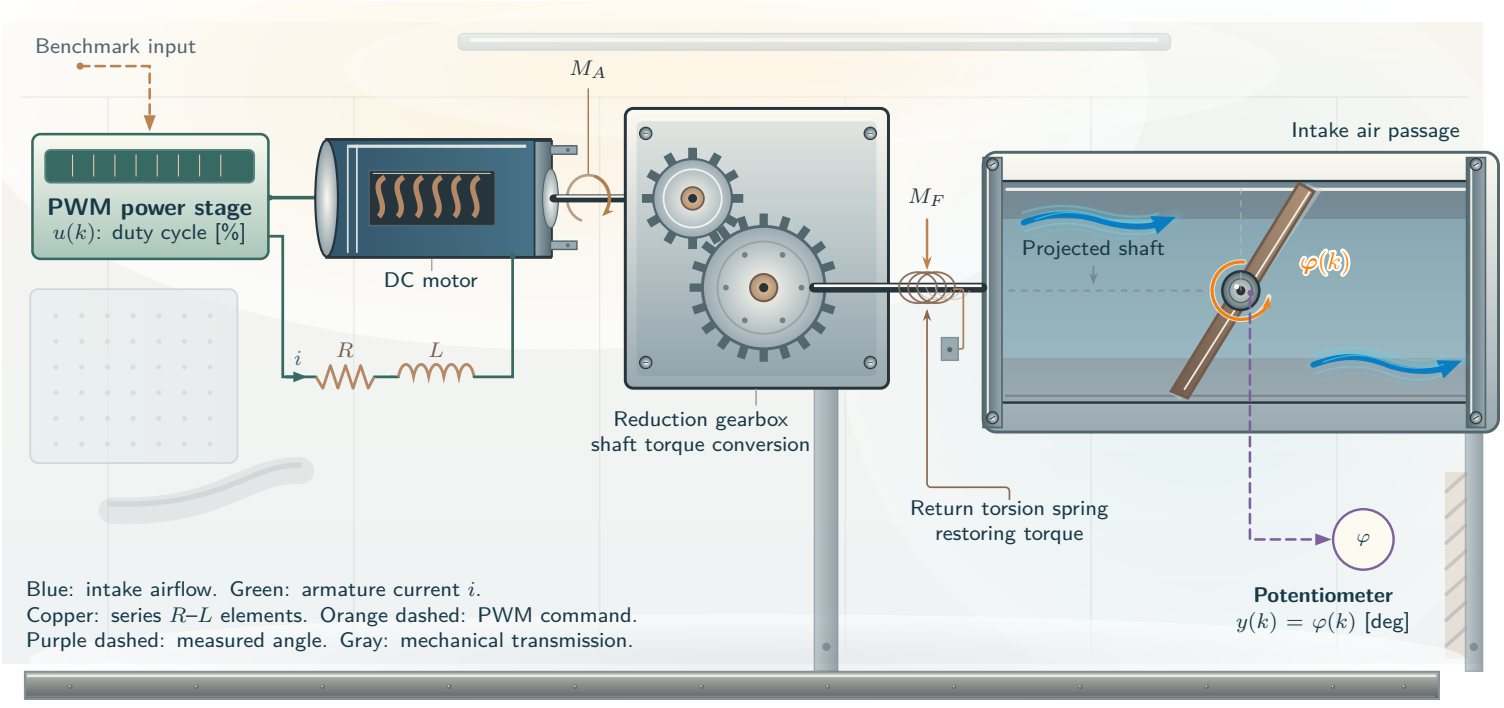
Model scope: Eq. (32) with explicit force saturation and slip friction. No rail-impact or static-sticking model is implied. The educational learning study may declare different friction settings in its case sheet.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

# Automotive throttle: actuation and identification

Paper section 3.4: *Electro-mechanical throttle*

## Physical chain: PWM drive, DC motor, gear box, spring and throttle plate




Automotive engine-test-cell rig (illustrative equipment and lighting; not to scale)

**Acquisition**
Sample interval: $T_0 = 10$ ms.

Input: PWM duty [%].
Output: angle [deg].

Published plate range:
$10^\circ$ closed to $90^\circ$ open.

**Identification challenge**
Friction, nonlinear spring and mechanical compliance can change the input–output response.

No calibrated internal parameters are inferred here.

**Training experiment**
Multisine excitation; $N = 10\,000$.
Fit dynamic models using the training record only. Keep measured duty and angle in their stated units.

**Independent validation**
Quasi-APRBS excitation; $N = 2\,500$.
Evaluate one-step and free-run prediction separately; report errors, residuals and model complexity.

**Repository evidence**
PWA/TS-inspired educational routines are available. Original measured time series have not been recovered; fixtures are synthetic.

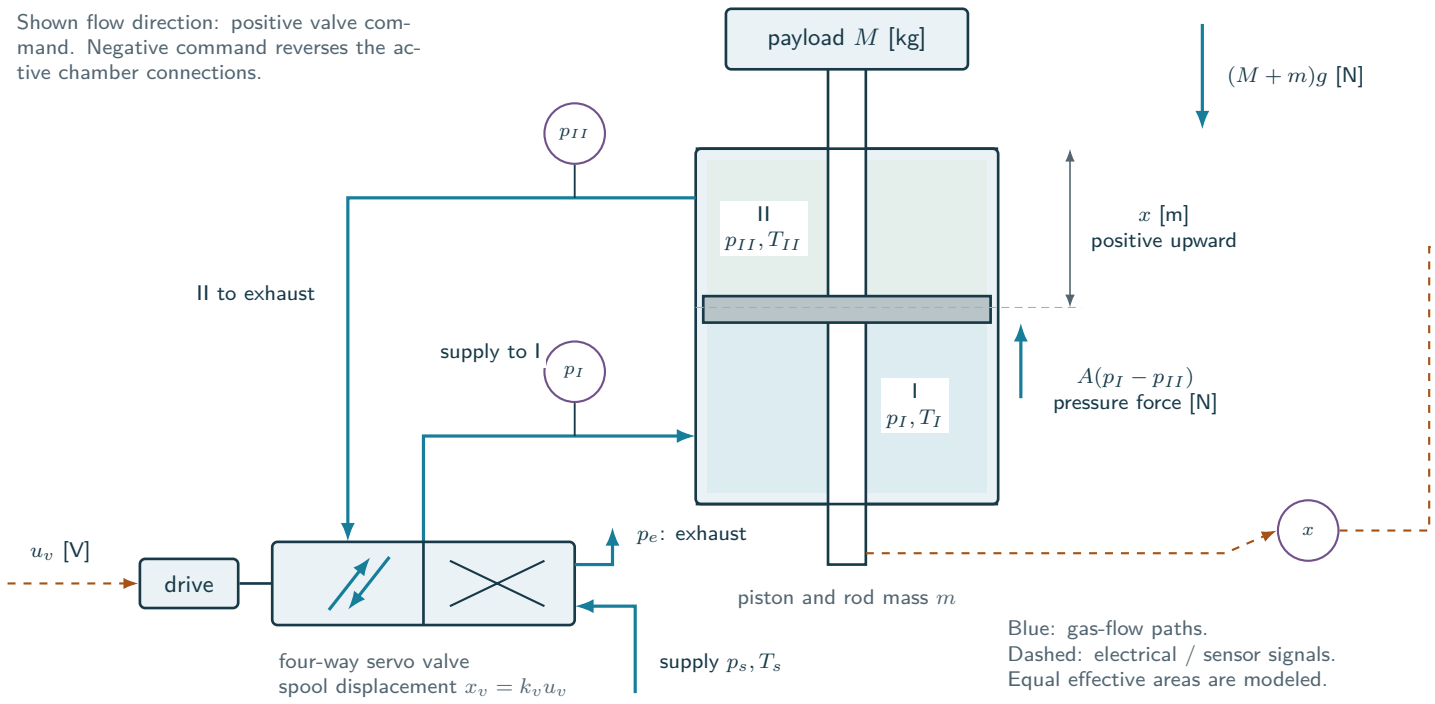
Technology illustration, not a calibrated throttle model or a production assembly drawing. Gear ratio, motor constants, spring law and friction are not inferred. See `THROTTLE_RECOVERY.md` for data access status.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

# Vertical servo-pneumatic actuator

Paper section 3.5: *Servo-pneumatic actuator*

## Two variable-volume chambers, valve flow and a vertical payload




**Absolute physical variables**

$p_I, p_{II}$: pressure [Pa].
$T_I, T_{II}$: temperature [K].
$m_I, m_{II}$: gas mass [kg].
$A$: effective piston area [m$^2$].

$V_I = V_0 + Ax$
$V_{II} = V_0 - Ax$
$p_j = m_j R T_j / V_j.$

**Physical variant state**

$[x, v, m_I, m_{II}, T_I, T_{II}]^\mathsf{T}$

Mass and energy balances evolve each chamber. Flow can choke or reverse. Valve and thermal parameters require calibration.

**Vertical force balance**

$(M+m)\dot{v} = A(p_I - p_{II})$
$\quad - f(v) - (M+m)g - F_{ext}.$

The physical variant includes gravity explicitly.

**Three model meanings**

Eqs. (33)–(34): literal survey form.
Local identified models: deviation states with operating offsets removed.
Six-state model: separate physical variant.

**Units and payload**

Mass-specific $R = R_u / M_{air}$ [J/(kg K)].
Each physical simulation has constant payload; changing payload in motion needs additional assumptions.

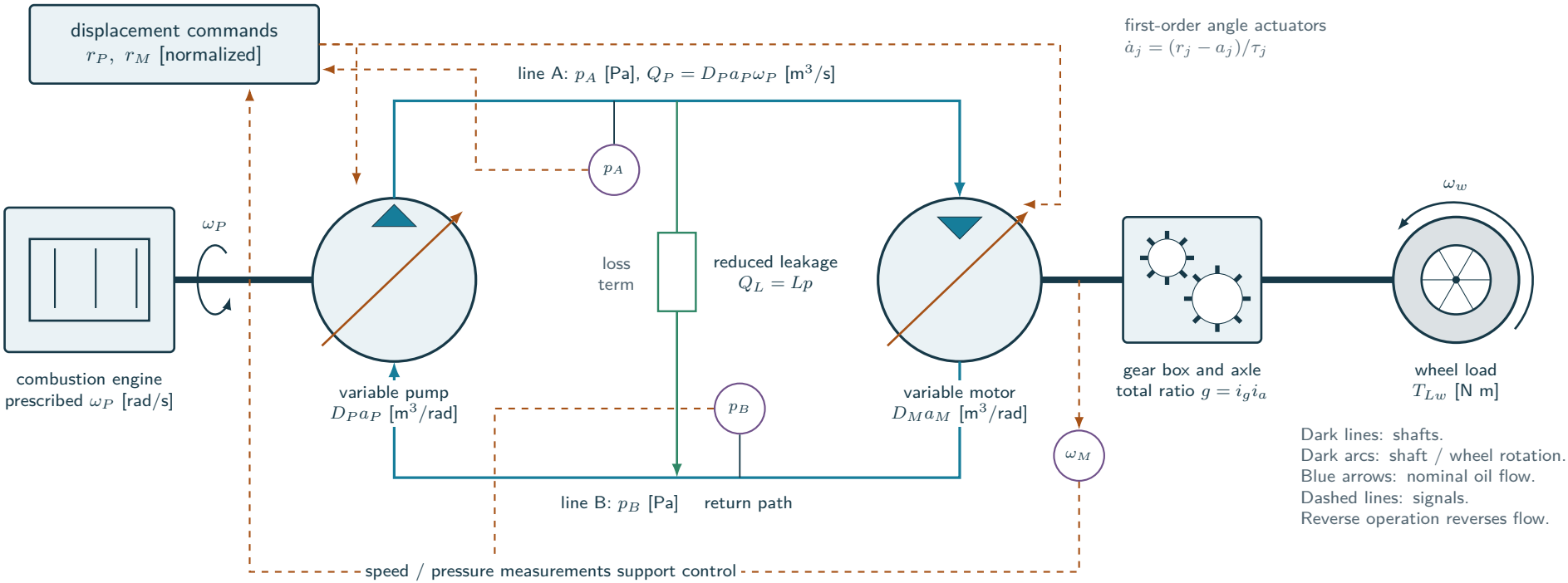
Functional explanatory diagram, not a standardized pneumatic schematic. The survey's gravity omission and gas-constant units are discussed in PNEUMATIC_AUDIT.md. The physical valve closure is an explicitly parameterized teaching variant, not an identified replica of the apparatus.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

# Hydrostatic drive: closed oil circuit and vehicle load

Paper section 3.6: *Hydrostatic transmission in mobile working machines*

## Variable-displacement pump and motor couple hydraulic pressure to shaft speed




**SI variant state and storage**

$[a_P, a_M, p, \omega_M]^\mathsf{T}$, $p = p_A - p_B$.
$C_\Delta \dot{p} = D_P a_P \omega_P - D_M a_M \omega_M - Lp$.
$C_\Delta$ [m$^3$/Pa]: effective capacitance.

**Mechanical balance**

$J\dot{\omega}_M = \eta D_M a_M p - b\omega_M - T_{Lw}/g$.
Motor torque uses motor displacement.
Displacement per revolution is converted through $D = V/(2\pi)$.

**Declared interpretation**

Eq. (35) remains a separate transcription. The SI variant requires explicit displacement and capacitance conventions. No electrical actuator calibration is inferred.

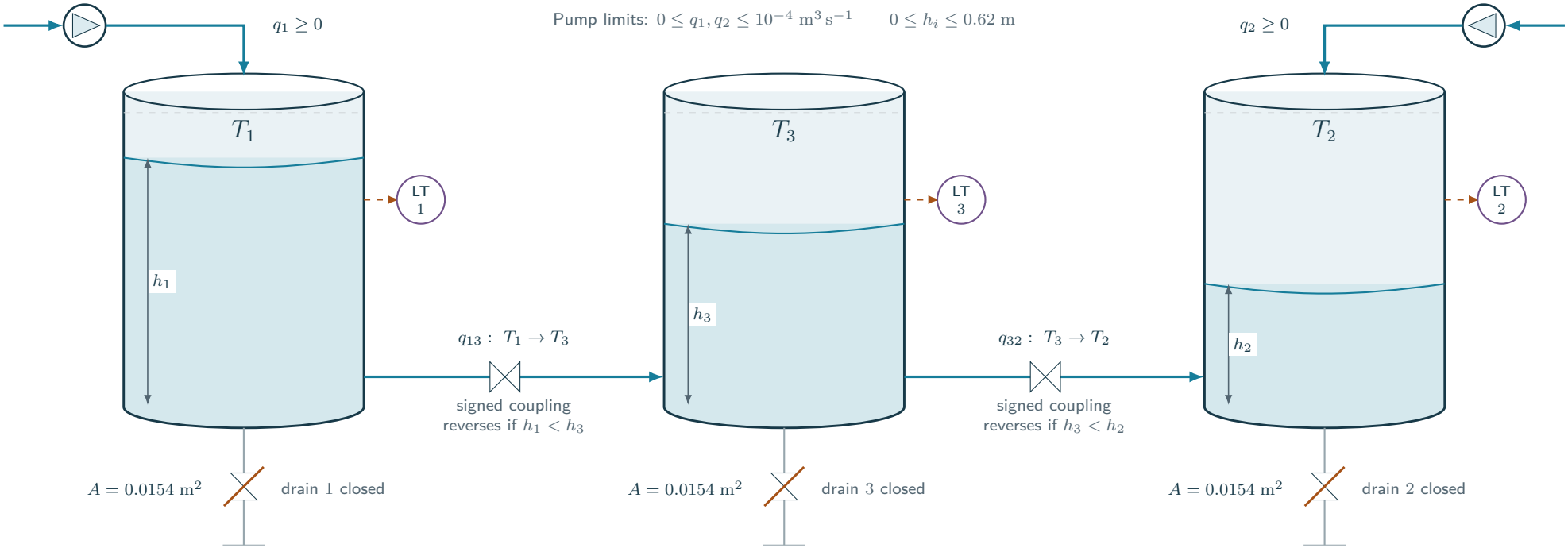
Functional circuit, not a standardized hydraulic drawing. Constant efficiencies are valid for the declared motoring study; no passive four-quadrant, cavitation or relief-valve model is implied. See HYDROSTATIC_SOURCE_REVIEW.md for source conflicts and unresolved calibration.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

# Three interconnected tanks

Paper section 3.7: *Three-tank-system*

Physical arrangement: $T_1$–$T_3$–$T_2$ | State ordering in the implementation: $[h_1, h_2, h_3]^\mathsf{T}$




Blue: liquid flows. Dashed orange: level observations. Crossed valves: zero nominal discharge; drain/fault experiments are explicit extensions.

**Hydraulic coupling**

$$q_{ij} = a\,\mu\,\mathrm{sgn}(h_i - h_j) \cdot \sqrt{2g|h_i - h_j|}$$

$a = 0.5\times10^{-4}\ \mathrm{m^2}$, $\mu = 0.48$.

**Nominal level balances**

$$A\dot{h}_1 = q_1 - q_{13}$$
$$A\dot{h}_3 = q_{13} - q_{32}$$
$$A\dot{h}_2 = q_2 + q_{32}$$

Time in seconds; levels in metres.

**Conservation and operating limit**

$$\frac{d}{dt}A(h_1 + h_2 + h_3) = q_1 + q_2.$$

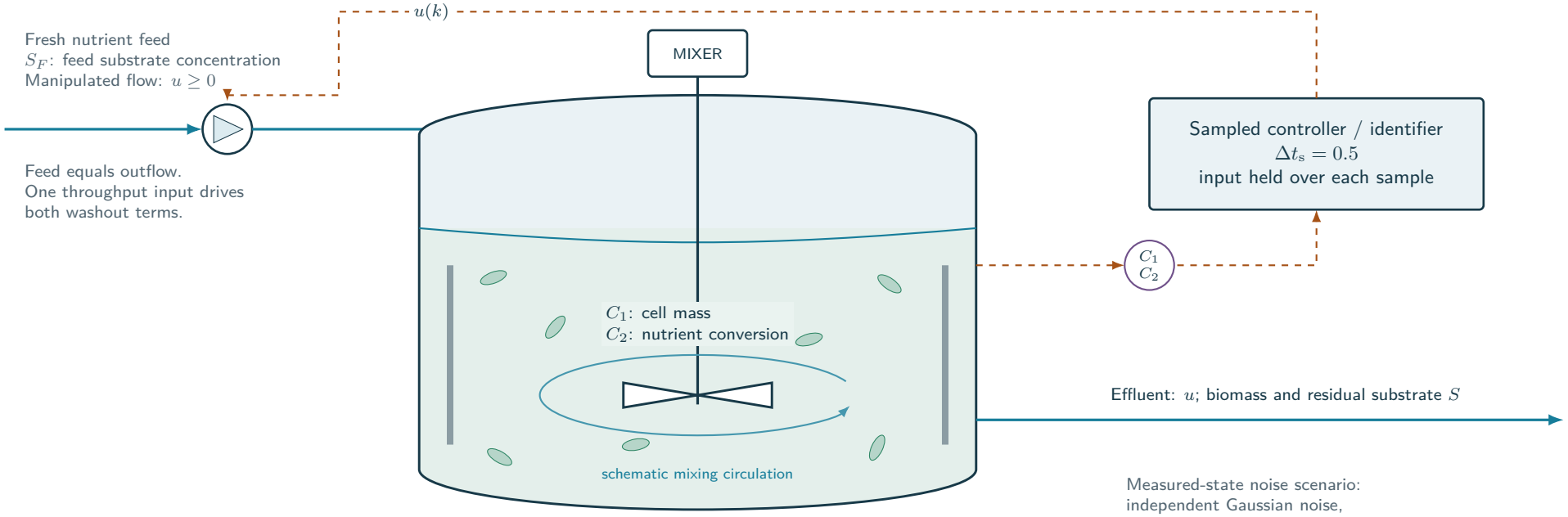
With closed drains, positive pumping rules out a steady equilibrium. Corrected area unit: $\mathrm{m^2}$.

Scope: three-state nominal model, Eq. (36). Vessel heights in this drawing are illustrative. A tracking experiment with continuous pumping requires an explicitly declared outlet extension.
Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

# Continuous biochemical reactor

Paper section 3.8: *Continuously operated bio-chemical reactor*

Two nonlinear states | Constant-volume, well-mixed process | Dimensionless model and time




**Normalized state definitions**

$C_2 = (S_F - S)/S_F, \quad 0 \leq C_1, C_2 \leq 1.$

$r = C_1(1 - C_2)\exp(C_2/\Gamma).$

Nominal $\Gamma = 0.4$, $\beta = 0.02$.

**Growth and washout balances**

$$\dot{C}_1 = -uC_1 + r$$

$$\dot{C}_2 = -uC_2 + r\frac{1+\beta}{1+\beta-C_2}.$$

No dimensional flow or time scale is supplied.

**What makes this a benchmark**

Stable and unstable operating regions; limit cycles and a non-unique input–yield relation.

One sampled transition uses **50 Euler substeps**, each of length $0.01$.

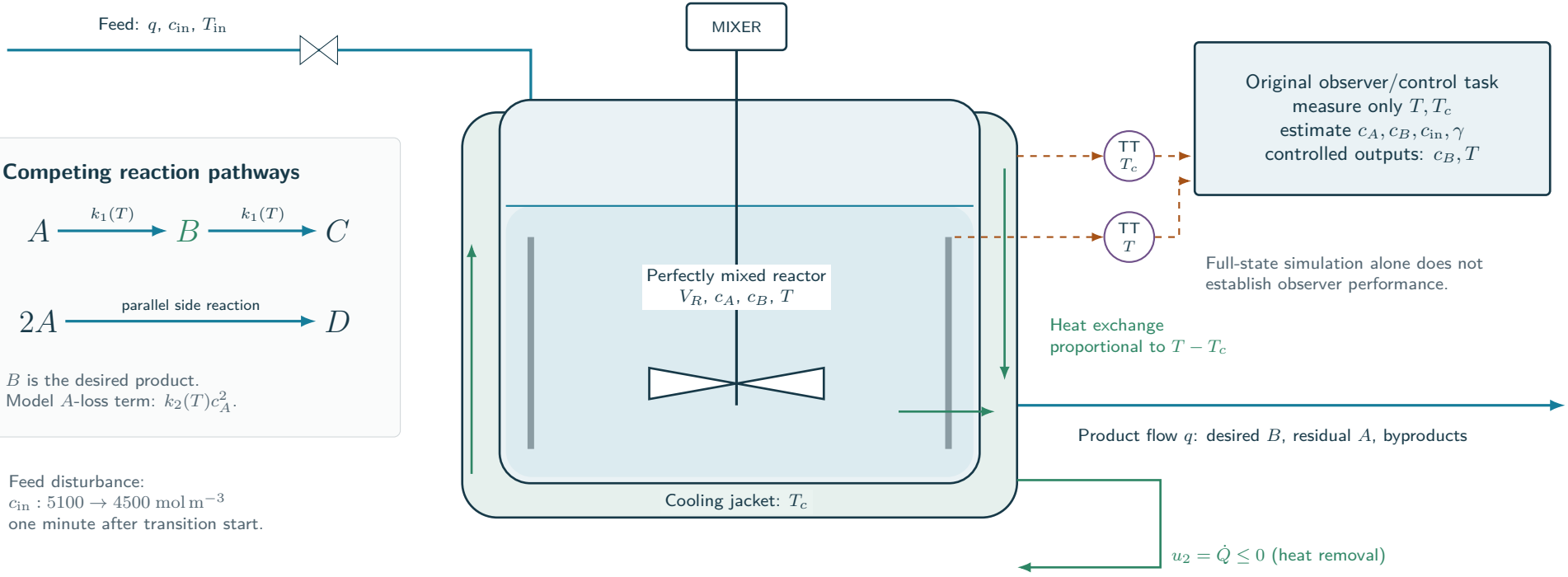
Scope: Eqs. (37)–(38), with dimensionless throughput $u$. The mixing hardware is explanatory. Equilibria must be recomputed from the equations; the printed biomass at $u = 0.769$ is not consistent with Eq. (37).

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

# Klatt–Engell jacketed CSTR

Paper section 3.9: *Continuously stirred chemical reactor*

Four states: $[c_A, c_B, T, T_c]^\mathsf{T}$ | Inputs: dilution $q/V_R$ and cooling capacity $\dot{Q}$ | Time: hours




### Competing reaction pathways

$$A \xrightarrow{k_1(T)} B \xrightarrow{k_1(T)} C$$

$$2A \xrightarrow{\text{parallel side reaction}} D$$

$B$ is the desired product.
Model $A$-loss term: $k_2(T)c_A^2$.

Feed disturbance:
$c_{\mathrm{in}} : 5100 \rightarrow 4500\ \mathrm{mol\,m^{-3}}$
one minute after transition start.

### Material balance and kinetics

$$\dot{c}_A = -k_1 c_A - k_2 c_A^2 + (c_{\mathrm{in}} - c_A)u_1$$
$$\dot{c}_B = k_1(c_A - c_B) - c_B u_1$$
$$k_i = k_{i0} \exp[-E_i/(T + 273.15)].$$

### Thermal balances

$$\dot{T} = h(c_A, c_B, T) + \alpha(T_c - T) + (T_{\mathrm{in}} - T)u_1$$
$$\dot{T}_c = \beta(T - T_c) + \gamma\,(1000 u_2).$$

$h$: reaction heat; 1000: MJ to kJ.

### Units and input constraints

$5 \leq u_1 \leq 35\ \mathrm{h^{-1}}$; $u_1 = q/V_R$.
$-8.5 \leq u_2 \leq 0\ \mathrm{MJ\,h^{-1}}$.

$c_A, c_B$ in $\mathrm{mol\,m^{-3}}$; $T, T_c$ in °C. Kelvin is used inside Arrhenius rates.

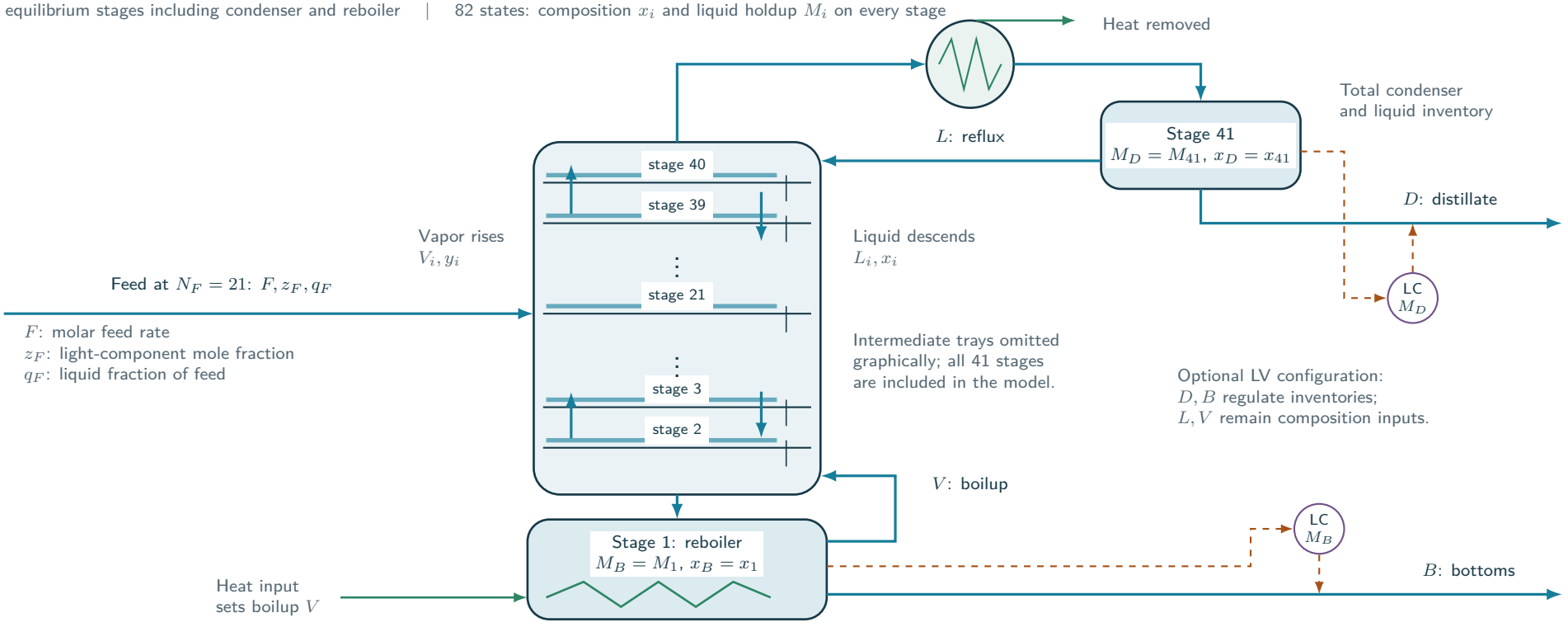
Scope: Eqs. (39)–(43). Celsius resolves the Table 7 temperature-label inconsistency with Eq. (41). The jacket flow arrows illustrate heat-transfer hardware; the model input is cooling capacity, not a specified coolant flow.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

# Column A: binary distillation

Paper section 3.10: *Continuously operated binary distillation column*

41 equilibrium stages including condenser and reboiler | 82 states: composition $x_i$ and liquid holdup $M_i$ on every stage




**Interior-stage conservation**

$$\dot{M}_i = L_{i+1} - L_i + V_{i-1} - V_i$$
$$\frac{d(M_i x_i)}{dt} = L_{i+1}x_{i+1} + V_{i-1}y_{i-1} - L_i x_i - V_i y_i.$$

Feed, condenser and reboiler use separate balances.

**Thermodynamics and hydraulics**

$y_i = \alpha x_i / [1 + (\alpha - 1)x_i]$, $\alpha = 1.5$.
$L_i = L_{0i} + (M_i - M_{0i})/\tau_l$ on trays.

Constant pressure; negligible vapor holdup; constant molar flows in each section.

**Interfaces and units**

$4 \times 4$: inputs $L, V, D, B$; outputs $x_D, x_B, M_D, M_B$.
Flows: kmol/min; holdups: kmol.

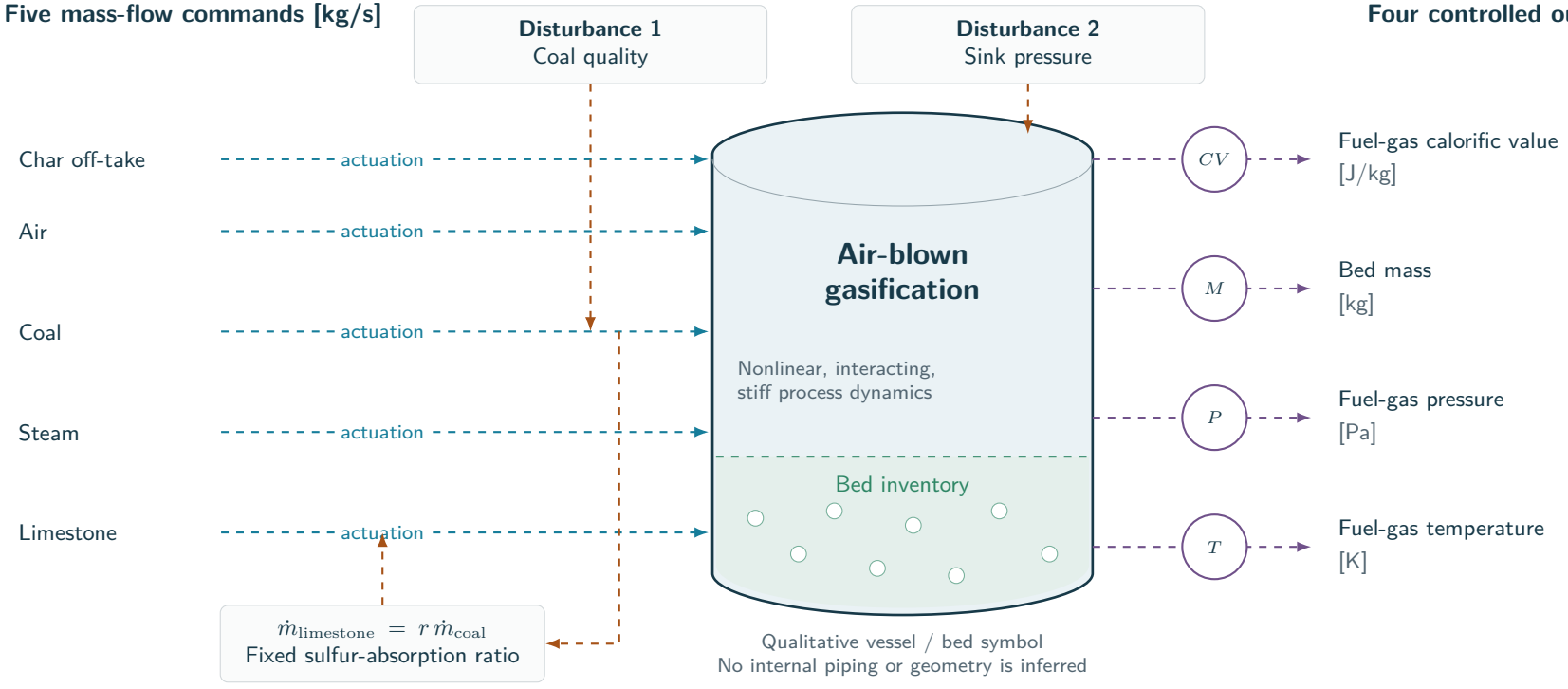
NTNU nominal values: $L_0 = 2.70629$, $V_0 = 3.20629$; $\tau_l = 0.063$ min.

Scope: Column A, Eqs. (47)–(54), aligned with Skogestad's NTNU reference code. Incoming vapor uses $V_{i-1}$, correcting the printed Eq. (47). Dashed D/B level loops describe the documented LV extension.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

# ALSTOM gasifier: nonlinear plant and control interface

Paper section 3.11: *ALSTOM Gasifier Control Benchmark II*




Four independent manipulated flows; the fifth follows coal. Magnitude and slew limits apply to the actuators.

**Official model family**

Nonlinear open-loop plant and multivariable PI baseline. The native package contains the dynamics; this drawing specifies the external signal interface.

**Challenge experiments**

Sink-pressure steps and harmonics; load and coal-quality changes. Compare maximum and integral absolute output errors with the declared bounds.

**Interpretation of results**

A completed simulation is not a passed control specification. The retained native PI results exceed the temperature band in all six pressure cases.

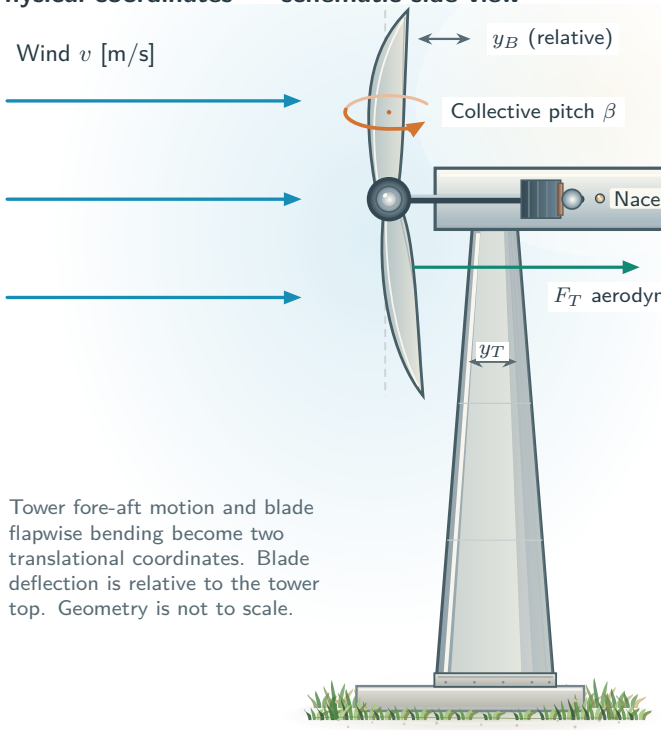
Dashed blue arrows: mass-flow control signals, not physical pipe directions; char is withdrawn from the process. Orange: disturbances and fixed-ratio linkage. Purple: output measurements, not material outlets. Vessel and bed are conceptual; this is not a construction P&ID.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

# Wind turbine: structure, drivetrain and aerodynamic coupling

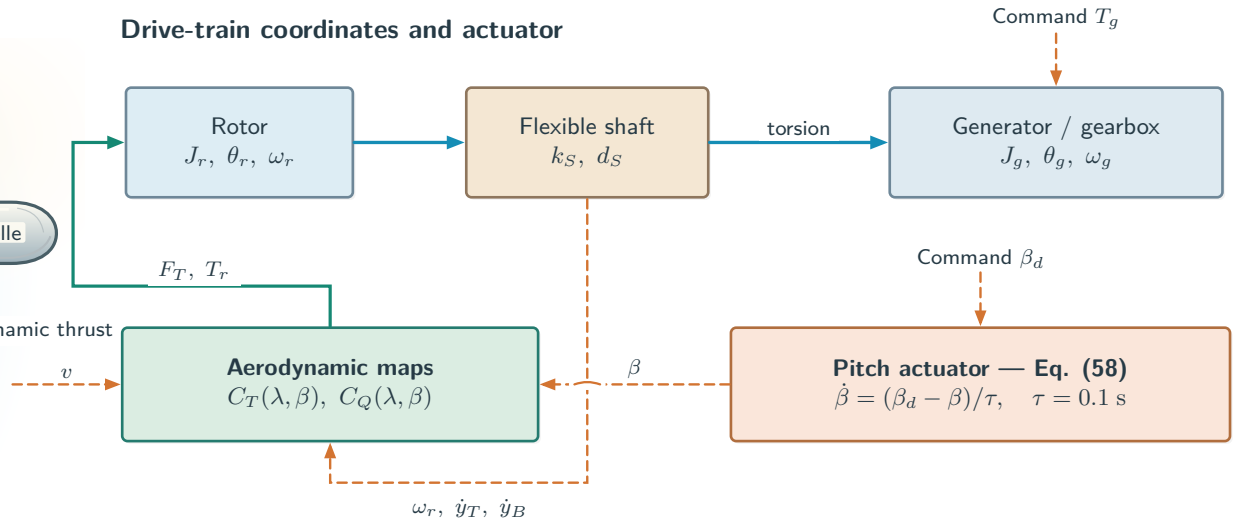
Paper section 3.12: *Wind turbine*

## Physical coordinates — schematic side view




Tower fore-aft motion and blade flapwise bending become two translational coordinates. Blade deflection is relative to the tower top. Geometry is not to scale.

## Drive-train coordinates and actuator




$$v_{\text{eff}} = v - \dot y_T - \dot y_B, \qquad \lambda = \frac{\omega_r R}{v_{\text{eff}}}, \quad R = 63\ \text{m}$$

$$F_T = \tfrac{1}{2}\rho\pi R^2 C_T v_{\text{eff}}^2, \qquad T_r = \tfrac{1}{2}\rho\pi R^3 C_Q v_{\text{eff}}^2$$

$$M\ddot q + D\dot q + Kq = [F_T, F_T, T_r, -T_g]^{\mathsf T}$$

Generator inertia, angle and torque are referred to the low-speed side. The gearbox ratio is absorbed into these quantities; gearbox dynamics are omitted.

### Nine-state reduced model

$q = [y_T, y_B, \theta_r, \theta_g]^{\mathsf T}$.
$x = [q^{\mathsf T}, \dot q^{\mathsf T}, \beta]^{\mathsf T}$.
Eqs. (55)–(58) retain four mechanical degrees of freedom and first-order pitch.

### Control-oriented repository variant

The bundled 2022 ROSCO rotor map supplies the aerodynamic coefficients. Local LQR/PDC gains are newly computed; they do not reproduce Table 12.

### High-order verification family

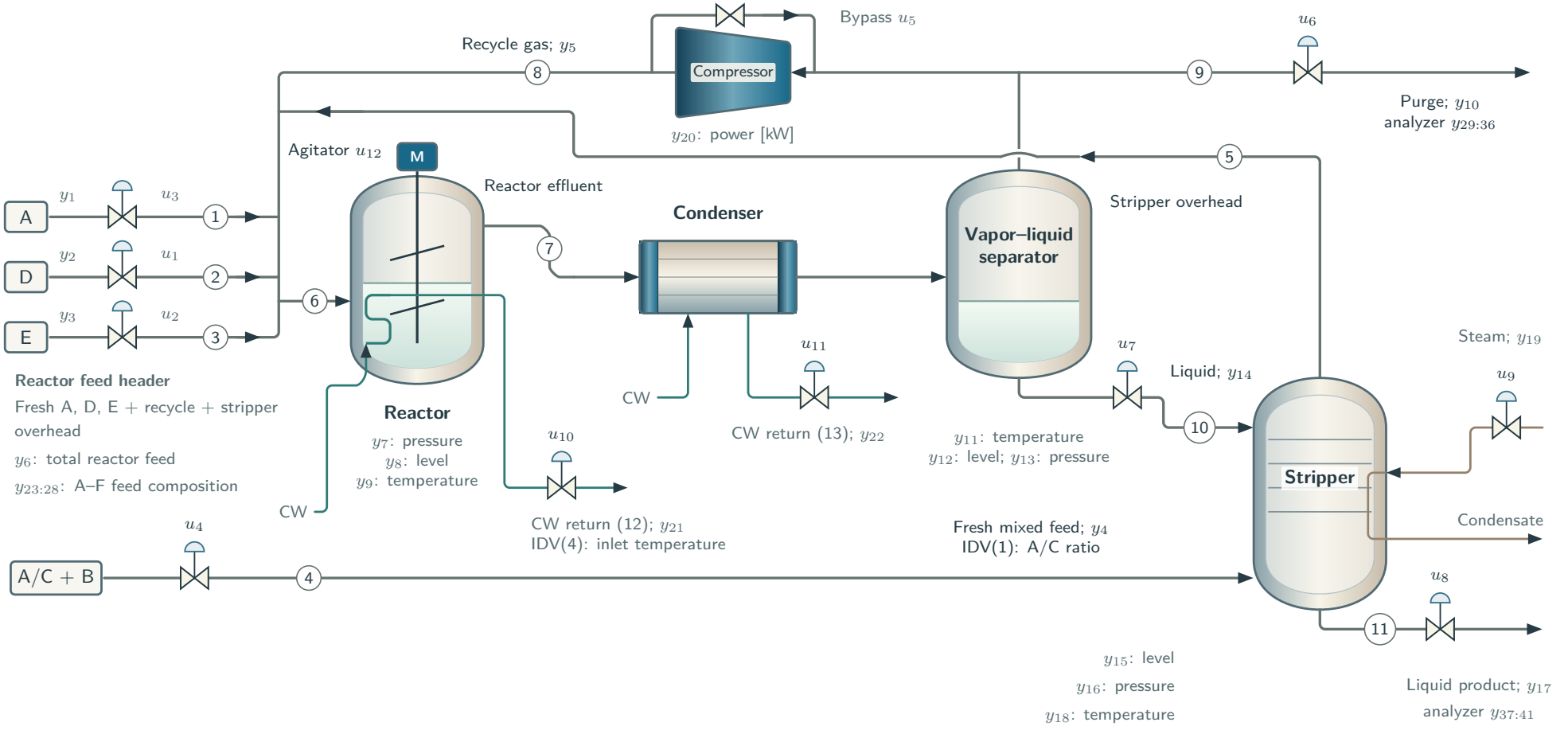
The article also names NREL FAST. The repository executes a later OpenFAST 5.0.0 NREL 5 MW case; it is distinct from this nine-state model.

Blue: mechanical propagation or incident wind. Green: aerodynamic force/torque. Solid orange arc: blade-axis pitch. Orange dashed: model signals and commands. The reduced plant has two commands $(T_g, \beta_d)$, wind disturbance $v$, and nine states. FAST/OpenFAST is a separate high-order verification model family.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

# Tennessee Eastman: process, utilities and instruments

Paper section 3.13: *Tennessee Eastman process control problem*




**Chemistry and objective**
A, C, D, E react; B is inert; G and H are products; F is a byproduct. Reject disturbances while preserving inventories, product rate and quality.

**Model boundary**
50 states, 12 commands, 41 measurements, six operating modes. The archived native C revision also defines `NX`=50. Downstream G/H separation is outside the benchmark.

**Reading the drawing**
Gray: process. Teal: CW. Tan: steam/condensate. T-junctions connect. White bridge: crossing. Finishes and fluid tints are illustrative. $u_i = \mathrm{XMV}(i)$; $y_i = \mathrm{XMEAS}(i)$.

Topology checked against Fig. 13 and the Ricker archive; signal indices checked against `temexd_mod.c`. Original explanatory flowsheet, not a construction P&ID. Instrument labels identify measurements, not a prescribed controller.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.

# Tennessee Eastman: benchmark signal reference

Paper section 3.13: *Tennessee Eastman process control problem*

**12 manipulated variables: XMV**

Valve and agitator commands are expressed as percentages.

| | |
|---|---|
| 1 | D feed, stream 2 |
| 2 | E feed, stream 3 |
| 3 | A feed, stream 1 |
| 4 | A/C mixed feed, stream 4 |
| 5 | Compressor recycle bypass |
| 6 | Purge, stream 9 |
| 7 | Separator liquid, stream 10 |
| 8 | Stripper product, stream 11 |
| 9 | Stripper steam |
| 10 | Reactor cooling-water outlet |
| 11 | Condenser cooling-water outlet |
| 12 | Reactor agitator command [%] |

Source terminology sometimes calls the condenser coolant the separator coolant; it is the same thermal path. Commands and realized valve positions are distinct model quantities.

**Representative disturbance paths**

IDV(1): A/C feed ratio, constant B.
IDV(4): reactor coolant inlet temperature.
IDV(6): loss of A feed.
IDV(13): kinetics drift.
IDV(14–15): coolant valve sticking.

The repository's archived native runs cover nominal operation, IDV(1) and IDV(4); this diagram adds no new execution evidence.

**Continuous measurements: XMEAS(1:22)**

| | | |
|---|---|---|
| 1 | A feed | kscmh |
| 2 | D feed | kg/h |
| 3 | E feed | kg/h |
| 4 | A/C mixed feed | kscmh |
| 5 | Recycle flow | kscmh |
| 6 | Reactor feed | kscmh |
| 7 | Reactor pressure | kPa gauge |
| 8 | Reactor level | % |
| 9 | Reactor temperature | °C |
| 10 | Purge flow | kscmh |
| 11 | Separator temperature | °C |
| 12 | Separator level | % |
| 13 | Separator pressure | kPa gauge |
| 14 | Separator underflow | $m^3/h$ |
| 15 | Stripper level | % |
| 16 | Stripper pressure | kPa gauge |
| 17 | Stripper underflow | $m^3/h$ |
| 18 | Stripper temperature | °C |
| 19 | Stripper steam | kg/h |
| 20 | Compressor power | kW |
| 21 | Reactor coolant outlet | °C |
| 22 | Condenser coolant outlet | °C |

kscmh: thousands of standard cubic meters per hour.

**Composition analyzers: XMEAS(23:41)**

All concentrations are molar percentages.

**Reactor feed, stream 6**
$23{:}28 \longleftrightarrow A, B, C, D, E, F$

**Purge, stream 9**
$29{:}36 \longleftrightarrow A, B, C, D, E, F, G, H$

**Product, stream 11**
$37{:}41 \longleftrightarrow D, E, F, G, H$

Analyzer sample/hold, noise and delay are part of the simulator. A composition signal is not an instantaneous complete state observation.

**Verify the implemented state count**

The survey describes 50 original states. The archived native C revision also uses `NX=50`, with 50 derivatives. Its introductory comment says 52, but that comment conflicts with the executable declarations. See the state-count audit; extra diagnostic outputs are not additional continuous states.

**From the flowsheet to a learning or diagnostic experiment**

Record the simulator revision, operating mode, solver, time base, random seed, initial state, command limits and disturbance schedule. Select measured channels before fitting or tuning. Respect the chronological split and analyzer timing.

Markov/HMM sensor modes in the course are explicitly synthetic overlays. They must not be identified with official process disturbances or with experimentally calibrated equipment-failure probabilities.

Signal labels and units checked against the Ricker/Bathelt archive's documented C interface. State count checked against its executable declarations. The gallery links the archive, source hash and state-count audit.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.